CSE 623 ML Project Report Week-6
Project Name: A Traditional Machine Learning Pipeline for Robust Wheat Counting in Multi-Object Field Scenes
Group Name: Gradient Descent

1. Objective

For this week our task was to introduce spatial localization into the pipeline using the sliding window approach. Also, to evaluate its impact on counting accuracy and comparing it with the global regression-based methods.

Additionally, our target was to analyze the performance of the other models such as XGBoost and Extra Trees before integrating localization.

2. Dataset & Sampling Strategy

The same sampling strategy from previous weeks was used:

- Total domains: 18
- Images per domain: 10
- Total images: 180

This ensured:

- Equal domain representation
- Reduced computational cost
- Consistency with previous experiments

The ground truth wheat head counts were derived from the annotations provided in the dataset.

3. Sliding Window Detection Pipeline

The sliding window approach is use to localize the wheat heads (which was a limitation as observed in Week-5), and we have implemented the sliding window approach.

Step 1 - Window Generation

- A fixed-size window which scans the whole image.
- Window generation is controlled using the window and stride size.

Step 2 - Patch Classification

- Every patch is then classified using trained models.
- The output is shown as whether the wheat head is present or absent.

Step 3 - Detection Aggregation

- All positive detections are counted.
- The number of detected patches are considered as the final wheat head count.

4. Evaluating New Models before applying Sliding Window:
We evaluated following regressors before the sliding window:

1. **XGBoost Regressor**

- Prediction Biasness gets optimized by sequential boosting
- Domain variation is handled properly then before

2. **Extra Trees Regressor**

- Higher Randomness gives lower Variances
- We get good complementary behavior to Random Forest

Observations:

- We got improved performance over both the model, RF and SVM.
- However they still operate at image level and lacks spatial awareness to that.

5. Obtain Results after Sliding Window:
Following are the key outcomes:

- We got a significant reduction in mean absolute error (MAE).
- The counting accuracy improved then before.
- Our prediction aligned closer to ground truth counts.

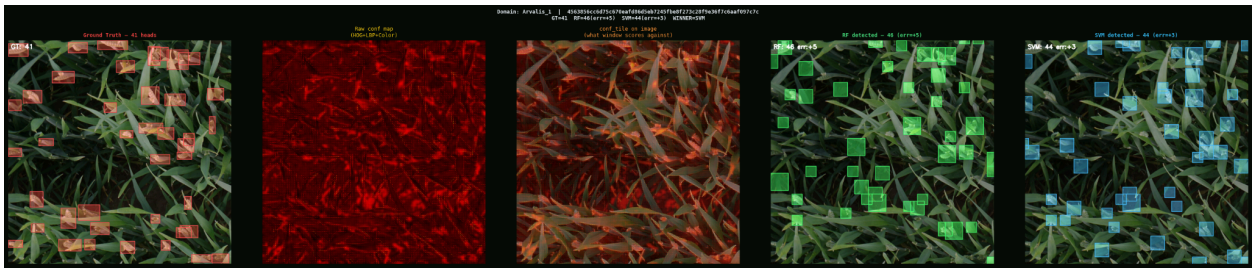
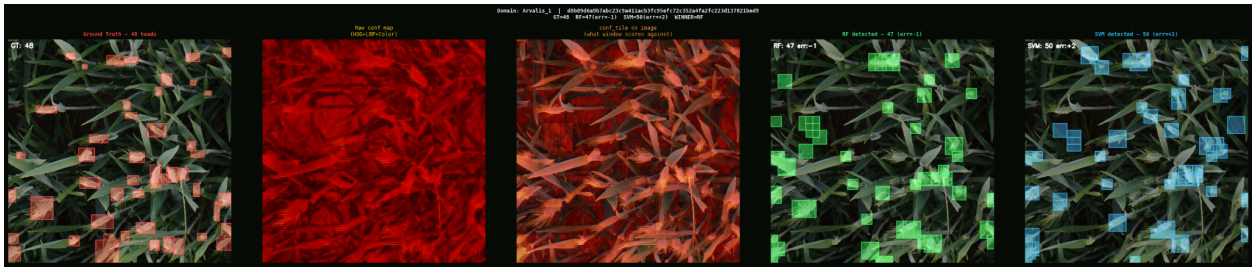
Important Observation:

- Localization (Bounding Boxes) were not that accurate as we thought them to be at first but still we can account for the significant improvements than previous outputs because the counting showed less errors.

6. Failure Analysis
Following are the observed Issues:

- Overlapping detections: Multiple windows were detecting the same wheat heads which were causing to give duplicate counting
- Imprecise localization: Location of bounding boxes was not found to be accurate.
- False positives: Background patches were identified as wheat heads which should not be the case.

7. Target For Next Week

- For next week, we will change different parameters on which sliding window is doing calculation of probability of whether to detect that particular patch was wheat or not.
    - Parameters like **combined** (hue score+val score+sat score), **w_spatial**, **diversity distance**, **confidence threshold.**

- We will see whether NMS alone is capable of separating overlapping wheat heads from each other or not.
- After all this, it will calculate localization accuracy as per its parameters and will identify the best parameters.